Using Hough transform, show that the following points are collinear. Also find the equation of the line. (1,2) (2,3) (3,4)

Ans

Equation of the line

$$\boxed{y = mx + c}$$

Step i)

To perform Hough transform, convert the line from (x, y) plane to (m, c) plane.

$$\boxed{c = -mx + y}$$

i) For (x, y) (1,2).

$c = -m(1) + (2)$

$c = -m + 2,$

if $c = 0 \Rightarrow m = 2$

if $m = 0 \Rightarrow c = 2.$

Then $(m, c) \Rightarrow (2, 2)$

ii) for (2,3) $c = -2m + 3.$

if $c = 0 \Rightarrow m = 3/2 = 1.5$

if $m = 0 \Rightarrow c = 3$

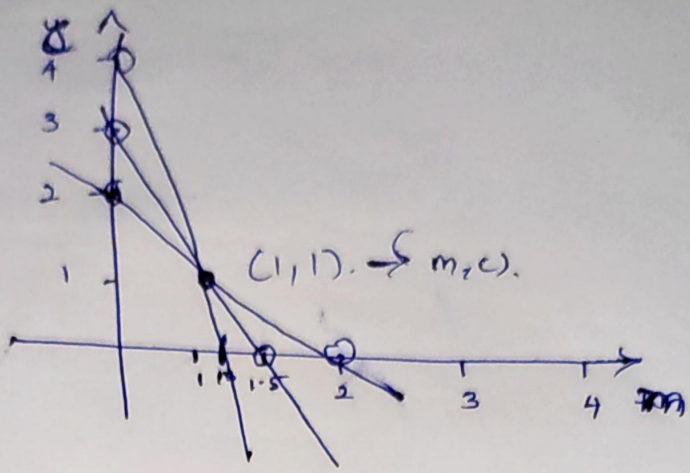
Then $(m, c) \Rightarrow (1.5, 3).$

iii) for (3,4) $c = -3m + 4.$

if $c = 0 \Rightarrow m = 4/3 = 1.33$

if $m = 0 \Rightarrow c = 4$

then $(1.33, 4)$



original Eqn/..

$$y = mx + c$$

$$\boxed{y = x + 1}$$ equation.

final check.

the point (1,2) $y = x + 1$

$y = 1 + 1$

$y = 2$

(2,3) $y = 2 + 1$

$y = 3.$

(3,4) $y = 3 + 1$

$y = 4$

Exercise

Using Hough transform show that the points (1,1) (2,2) and (3,3) are colinear and find the equation. $y = x$